



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

SCORE-3.0

15-07-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Main)

TEST DATE : 15-07-2021

ANSWER KEY

PHYSICS

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-1	A.	C	C	C	A	C	C	A	A	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	C	C	A	B	A	A	A	A
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.10	12.00	2.00	0.25	2.00	4.00	2.00	14.00	4.00	16.00

CHEMISTRY

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-1	A.	B	A	B	B	C	A	C	C	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	C	B	D	C	C	C	B	A
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	28.00	1.83	3.00	12.75	3.38	5.00	-4.00	-3.00	6.00	26.80

MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-1	A.	B	D	C	A	A	C	A	D	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	A	D	B	D	B	D	C	A
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.50	0.33	2.00	2.00	34.64	2.50	2.00	0.66 or 0.67	8.66 or 8.67	50.00

HINT – SHEET

PHYSICS

SEC-1

1. **Ans (C)**

Value of masses after explosion

$$\frac{2m}{5}, \frac{2m}{5}, \frac{m}{5}$$

$$P_i = P_f$$

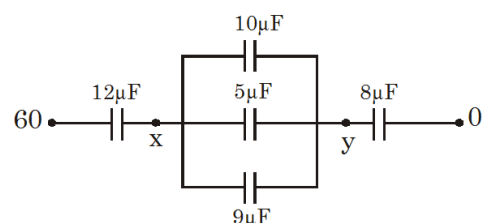
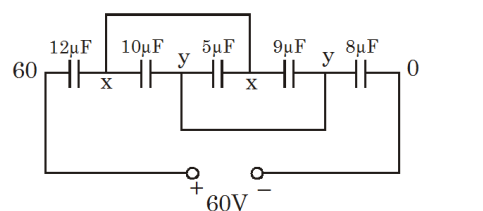
$$0 = \frac{2m}{5}\hat{v}_i + \frac{2m}{5}\hat{v}_j + \frac{m}{5}\hat{v}'$$

$$\hat{v}' = -2\hat{v}_i - \hat{v}_j$$

$$|\hat{v}'| = v\sqrt{5}$$

2. **Ans (C)**

Theory based

4. **Ans (A)**

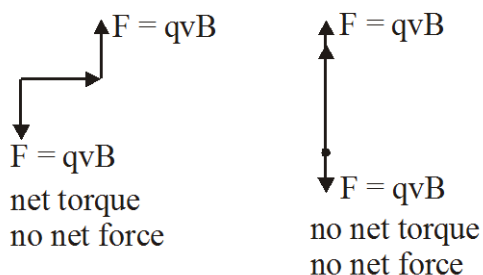
$$\frac{1}{C_{eq}} = \frac{1}{12} + \frac{1}{24} + \frac{1}{8}$$

$$C_{eq} = 4\mu F$$

$$q_{5\mu F} = \frac{240}{24} \times 5 = 50\mu C$$

5. **Ans (C)**

Since q_{net} on dipole = 0 $\Rightarrow$ no net force



6. **Ans (C)**

As shown, we conclude that A and C are analogue signals but B is digital signal.

7. **Ans (A)**

From A : $\Delta Q > 0$, $\Delta U = 0$, $\Delta W > 0$

From B : $\Delta Q < 0$, $\Delta U = 0$, $\Delta W < 0$

8. **Ans (A)**

The number for yellow colour is 4. The number for violet colour is 7. Brown colour gives a multiplier of 10^1 . Gold indicates a tolerance of 5%. So the resistance of the given resistor is $47 \times 10\Omega \pm 5\%$

(2) In the second code the body of the resistor one, end of a resistor and a dot on the body of the resistor carry specific colours.

The colour of the body gives the first significant figure.

The colour of the end gives the second significant figure.

The colour of the dot on the body of the resistor gives the decimal multiplier i.e., the number of zeros that will follow the two significant figures.

The colour of the ring R indicates the percentage accuracy of resistance or tolerance limits of the value of resistance.

9. **Ans (C)**

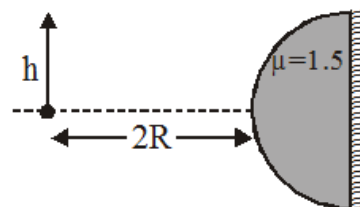
$$f = R/2$$

10. **Ans (D)**

Here, three optical phenomena take place— first refraction, then reflection, and finally refraction.

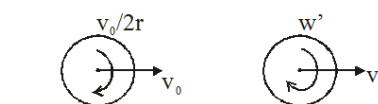
For refraction at 1st

$$\frac{1.5}{v} - \frac{1}{-2R} = \frac{1.5 - 1}{R} \Rightarrow \frac{1.5}{v} = 0 \Rightarrow v = \infty$$



i.e., rays after refraction get parallel to axis and strike the mirror normally, get retraced back and the final image is formed at the same place where the object is and of the same size. Image would be real.

11. **Ans (A)**



Angular momentum can be conserved about lowest point

$$mv_0r + \frac{2}{5}mr^2\frac{v_0}{2r} = mv'r + \frac{2}{5}mr^2\frac{v'}{r}$$

$$\frac{10mv_0r + 2Mv_0r}{10} = \frac{7mv'r}{5}$$

$$v' = \frac{6v_0}{7}$$

13. **Ans (C)**

Let initially they have n atoms

A $\rightarrow$ the half time two hours

B $\rightarrow$ the half time four hours

After two hours A possesses 1 half life and B

possesses $\frac{1}{2}$ half life

$$\text{Remaining atoms of A} = \frac{x}{2^1}$$

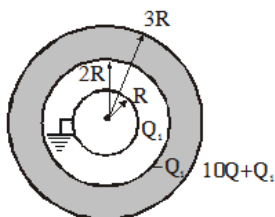
$$\text{Remaining atoms of B} = \frac{n}{(2)^{\frac{1}{2}}}$$

14. **Ans (C)**

$$\frac{k_1(T_1 - T)}{\ell_1} = \frac{k_2(T - T_2)}{\ell_2}$$

15. **Ans (A)**

$$\frac{KQ}{R} - \frac{KQ_1}{2R} + \frac{K(10Q + Q_1)}{3R} = 0$$



$$3Q_1 + 20Q + 2Q_1 = 0 \Rightarrow Q_1 = -4Q$$

16. **Ans (B)**

$W_A > W_B$ as mass of water in A is more than in B

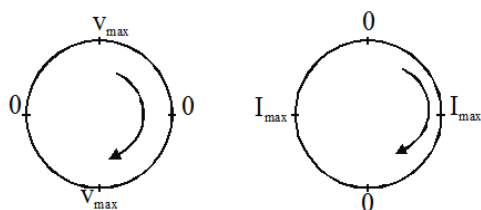
$$P_A = P_B$$

Area of A = Area of B

$$\text{or } P_A \text{ Area}_A = P_B \text{ Area}_B$$

$$\text{or } F_A = F_B$$

17. **Ans (A)**



$\Rightarrow$ Current lags voltage by 90°

18. **Ans (A)**

$$\text{Path difference } \Delta x = 2nt - \frac{\lambda}{2}$$

$$\text{For maximum } \frac{8}{3}t - \frac{\lambda}{2} = n\lambda$$

$$\frac{8}{3}t = \left(n + \frac{1}{2}\right) 630 \quad \dots(1)$$

$$\text{For minimum } \frac{8}{3}t - \frac{\lambda}{2} = \left(m - \frac{1}{2}\right) \lambda$$

$$\frac{8}{3}t = m(525) \quad \dots(2)$$

$$(1) + (2) \frac{n + \frac{1}{2}}{m} = \frac{2.5}{3}$$

$$\text{i.e. } n + \frac{1}{2} = 2.5$$

So from (1) $t = 590.625 \text{ nm}$

19. **Ans (A)**

At the open end pressure node is present. So that

at $x = 0$ P should be zero and $v = f\lambda$,

$$\text{for first overtone } \frac{3\lambda}{4} = 0.80$$

$$\Rightarrow f = 300$$

$$\Rightarrow \omega = 2\pi f = 600 \text{ t}$$

$$\text{Hence } P = A \sin \frac{15\pi}{8} x \cos 600 t$$

20. **Ans (A)**

At point P slope is $\tan 150^\circ$

$$\therefore \left(\frac{dp}{dT}\right) = \tan 150^\circ = \frac{-1}{\sqrt{3}} \quad \dots(1)$$

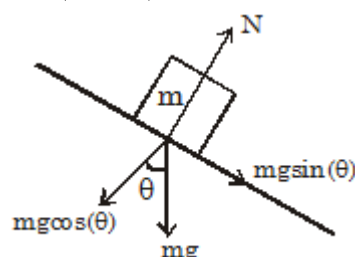
$$\text{Since } \rho = \frac{PM}{RT} \Rightarrow \frac{dp}{dT} = -\frac{PM}{RT^2} \quad \dots(2)$$

$$\text{Hence pressure } P = \frac{RT^2}{\sqrt{3}M}$$

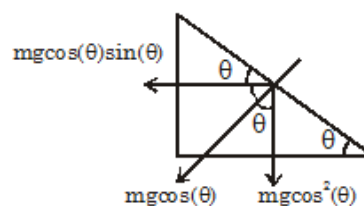
PHYSICS

SEC-2

2. **Ans (12.00)**



$$\Rightarrow N = mg \cos(\theta)$$



$$F_s = mg \cos(\theta) \sin(\theta)$$

$$= 25 \times \frac{4}{5} \times \frac{3}{5}$$

3. **Ans (2.00)**

$$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^2$$

$$R = 10 r$$

$$V = \frac{k(1000q)}{10r}$$

4. Ans (0.25)

$$y = \frac{x^2}{2} \Rightarrow \frac{dy}{dt} = \frac{1}{2} [2x] \times \frac{dx}{dt}$$

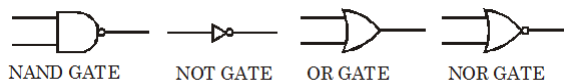
$$v_y = x \cdot v_x$$

$$x = \frac{t^2}{2} \text{ at } t = 2 \text{ sec} \Rightarrow x = 2$$

$$v_x = \frac{1}{2} \times 2t = 2$$

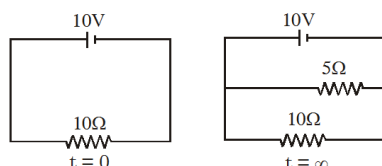
$$\vec{v} = 2\hat{i} + 4\hat{j} \text{ m/s}$$

5. Ans (2.00)



6. Ans (4.00)

At $t = 0$ 'L' behaves as open circuit at $t = \infty$ as short circuit



7. Ans (2.00)

Since particle executes SHM inside the tunnel

$$a = -\omega^2 x \quad \omega = \sqrt{\frac{GM}{R^2}}$$

$$\int_{\frac{GM}{R}}^V V dV = -\omega^2 \int_R^0 x dx$$

$$V^2 - \frac{GM}{R} = \omega^2 R^2$$

$$\therefore V = \sqrt{\frac{2GM}{R}}$$

$$\therefore (2)$$

8. Ans (14.00)

$$KE_{\max_1} = \frac{hc}{\lambda_1} - \phi$$

$$\text{Now } \phi = \frac{KE_{\max_1}}{3} = 2\text{eV (given)}$$

$$\text{So } \frac{hc}{\lambda_1} = 6 + 2 = 8\text{eV}$$

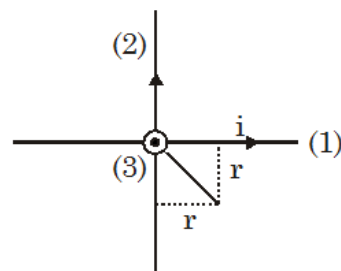
$$\text{and } \lambda_1 T_1 = \lambda_2 T_2$$

$$\text{So } \frac{hc}{\lambda_2} = 2 \frac{hc}{\lambda_1} = 16\text{eV}$$

$$KE_{\max_2} = \frac{hc}{\lambda_2} - \phi$$

$$KE_{\max_2} = 16 - 2 = 14\text{eV}$$

10. Ans (16.00)



Magnetic field at A due to

$$\text{Due to (1)} B_1 = \frac{\mu_0 i}{2\pi r} (-\hat{k})$$

$$\text{Due to (2)} B_2 = \frac{\mu_0 i}{2\pi r} (-\hat{k})$$

$$\text{Due to (3)} B_3 = \frac{\mu_0 i}{2\sqrt{2}\pi r} (\cos \theta \hat{i} + \sin \theta \hat{j})$$

$$\frac{\mu_0 i}{4\pi r} \hat{i} + \frac{\mu_0 i}{4\pi r} \hat{j}$$

$$B = B_1 + B_2 + B_3 = \frac{\mu_0 i}{4\pi r} \hat{i} + \frac{\mu_0 i}{4\pi r} \hat{j} - \frac{\mu_0 i}{\pi r} \hat{k}$$

CHEMISTRY

SEC-2

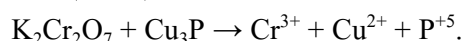
1. Ans (28.00)

$$N_1 V_1 = N_2 V_2$$

$$\left(\frac{63 \times 1.5 \times 10}{63} \right) \times 1 \times 2 = 1 \times (V + 2)$$

$$\Rightarrow V = 28 \text{ ml}$$

2. Ans (1.83)



$$\text{Moles} \times 6 = 1 \times (3 + 8) \Rightarrow \frac{11}{6} = 1.83$$

4. Ans (12.75)

$$\frac{r_1}{r_2} = \frac{1}{4}$$

$$\Delta E = E_2 - E_1 = 10.2 \text{ eV}$$

(x)

&

$$\frac{r_2}{r_4} = \frac{1}{4}$$

$$\Delta E = E_4 - E_2 = 2.55 \text{ eV}$$

(y)

$$(x + y) = 12.75 \text{ eV}$$

5. Ans (3.38)

$$\frac{a}{Rb} / \frac{8a}{27Rb} = \frac{27}{8} = 3.375 \simeq 3.38$$

7. **Ans (-4.00)**

$$K_{eq} = 1 \times 10^{-5}$$

$$\Rightarrow x = 1 \quad y = -5$$

$$(x + y) = 1 - 5 = -4$$

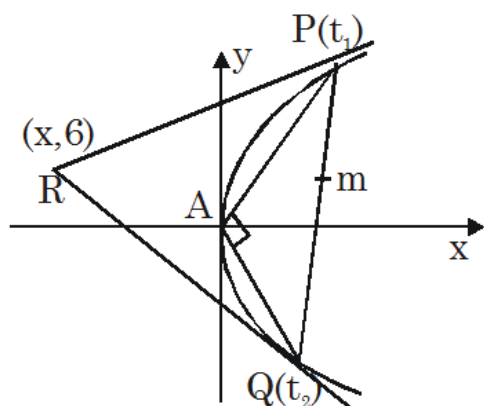
9. **Ans (6.00)**

Ba belongs to 6th period

MATHEMATICS

SEC-1

1. **Ans (B)**



$$a = 2$$

$$R(at_1t_2, a(t_1 + t_2)) = (x, 6)$$

$$t_1 + t_2 = 3 \quad \dots(i)$$

$$AP \perp AQ \Rightarrow t_1t_2 = -4 \quad \dots(ii)$$

$$\text{from (i) and (ii)} \Rightarrow t_1^2 - 3t_1 - 4 = 0$$

$$t_1 = 4, t_2 = -1$$

$$\text{point } P(32, 16) \quad Q(2, -4)$$

$$\text{mid point of } PQ \text{ is } M(17, 6)$$

2. **Ans (D)**

$$AM \geq GM$$

$$\frac{64\sin^6\alpha + 729\cos^6\beta + 1 + 1 + 1 + 1}{6} \geq 6\sin\alpha\cos\beta$$

$$AM = GM \Rightarrow \sin\alpha = \frac{1}{2}, \cos\beta = \frac{1}{3}$$

3. **Ans (C)**

A $\rightarrow$ B is false if A is true and B is false so for
 $p \rightarrow (\sim p \vee \sim q)$ p must be true and $\sim p \vee \sim q$.

If false

4. **Ans (A)**

$$3^x = \tan\theta, x = -\frac{1}{2} \quad \theta = \frac{\pi}{6}$$

$$f(x) = \sin^{-1}(\sin 2\theta)$$

$$f(x) = 2\tan^{-1}(3^x) \text{ for } x = -\frac{1}{2}$$

$$f'(x) = \frac{2}{1 + (3^x)^2} 3^x \ln 3$$

$$f'\left(-\frac{1}{2}\right) = \frac{1}{2} \sqrt{3} \ln 3$$

5. **Ans (A)**

$$f'(x) = \frac{1}{4} [1 + (x-2)^2 + (x-2)^4 + \dots \infty \text{ terms}]$$

$$f'(x) = \frac{1}{4} \left[\frac{1}{1 - (x-2)^2} \right] = \frac{1}{4(x-1)(3-x)}$$

6. **Ans (C)**

$$\begin{vmatrix} 0 & \cos x & -\sin x \\ \sin x & 0 & \cos x \\ \cos x & \sin x & 0 \end{vmatrix} = 0$$

$$\Rightarrow \sin^3 x = \cos^3 x \Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\sum \tan\left(\frac{\pi}{3} + x\right) = \tan\left(\frac{\pi}{3} + \frac{\pi}{4}\right) + \tan\left(\frac{\pi}{3} + \frac{5\pi}{4}\right)$$

$$= 2 \tan\left(\frac{7\pi}{12}\right) = 2 \tan(105^\circ)$$

$$= -2(2 + \sqrt{3}) = -(4 + 2\sqrt{3})$$

7. **Ans (A)**

$$\frac{dy}{dx} = \frac{y^3}{e^{2x} + y^2}$$

$$e^{-2x} \frac{dx}{dy} = \frac{1}{y^3} + \frac{e^{-2x}}{y}$$

$$e^{-2x} = t$$

$$\frac{dt}{dy} + \frac{2t}{y} = -\frac{2}{y^3}$$

$$\text{I.F.} = e^{\int \frac{2}{y} dy} = y^2$$

$$\Rightarrow t \cdot y^2 = \int -\frac{2}{y} dy$$

$$e^{-2x} y^2 = -2 \ln y + c$$

$$y(0) = 1 \Rightarrow c = 1$$

$$y^2 = -2e^{2x} \ln y + e^{2x}$$

8. **Ans (D)**

$$\int_0^1 \tan^{-1} \left(\frac{\tan^2 x - 2}{3 \tan x} \right) dx$$

$$= \int_0^1 \left[\tan^{-1} \left(\frac{\tan x}{2} \right) - \tan^{-1} (\cot x) \right] dx$$

$$= k - \int_0^1 \left(\frac{\pi}{2} - x \right) dx = k - \frac{\pi}{2} + \frac{1}{2}$$

9. **Ans (A)**

$$x^2 + \left(y - \frac{5}{3} \right)^2 = 1$$

$$y^2 = x^3, P(t^2, t^3)$$

Normal at point P is

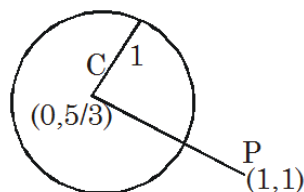
$$y - t^3 = -\frac{2}{3t}(x - t^2)$$

it passes through $\left(0, \frac{5}{3} \right)$

$$3t^3 + 2t - 5 = 0$$

$$(t-1)(3t^2 + 3t + 5) = 0$$

$$D < 0$$



$$\text{shortest distance} = CP - 1$$

$$= \frac{\sqrt{13}}{3} - 1$$

10. **Ans (C)**

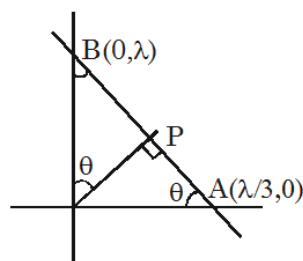
$$\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$$

$$x+1 = t^2 \Rightarrow dx = 2t dt$$

$$= \int \frac{(1+t^2) 2t dt}{(t^4+t^2+1)} = 2 \int \frac{\left(1 + \frac{1}{t^2}\right) dt}{\left(t - \frac{1}{t}\right)^2 + 3} = 2 \int \frac{dv}{v^2 + 3}$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t^2 - 1}{\sqrt{3}t} \right) + c = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}(x+1)} \right) + c$$

11. **Ans (A)**



$$\tan \theta = \frac{\lambda}{\left(\frac{\lambda}{3}\right)} = 3$$

$$\frac{BP}{AP} = \frac{\lambda \sin \theta}{\left(\frac{\lambda}{3}\right) \cos \theta} = 3 \tan \theta = 9$$

12. **Ans (D)**

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

Equation of normal at hyperbola

$$3x \cos \theta + 2y \cot \theta = 13$$

$$A \left(\frac{13}{3} \sec \theta, 0 \right), B \left(0, \frac{13}{2} \tan \theta \right)$$

$$O(0,0) \quad P(h,k)$$

$$h = \frac{13}{3} \sec \theta, k = \frac{13}{2} \tan \theta$$

$$\sec^2 \theta - \tan^2 \theta = 1 \Rightarrow 9x^2 - 2y^2 = 169$$

13. **Ans (A)**

Four digit numbers contains exactly two distinct digit

$$= \underbrace{{}^9C_2 (2^4 - 2)}_{\text{when 'O' is not selected}} + \underbrace{{}^9C_1 (2^3 - 1)}_{\text{when 'O' is selected}} = 567$$

14. **Ans (D)**

$$np = 4 \quad npq = \frac{4}{3}$$

$$q = \frac{1}{3}, p = \frac{2}{3}, n = 6$$

P(At least two success)

$$= 1 - p(\text{no. success}) - p(1 \text{ success})$$

$$= 1 - \left(\frac{1}{3} \right)^6 - {}^6C_1 \left(\frac{1}{3} \right)^5 \frac{2}{3}$$

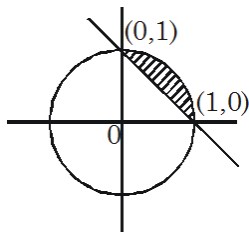
$$= 1 - \frac{13}{729} = \frac{716}{729}$$

15. Ans (B)

$$\sum_{r=3}^{100} \frac{r}{(r-2)! [1 + (r-1) + (r-1)r]} = \sum_{r=3}^{100} \frac{1}{r \cdot (r-2)!}$$

$$= \sum_{r=3}^{100} \left[\frac{1}{(r-1)!} - \frac{1}{r!} \right] = \frac{1}{2!} - \frac{1}{100!}$$

16. Ans (D)



$$\text{Area} = \frac{\pi}{4} - \frac{1}{2}$$

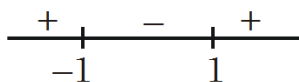
18. Ans (D)

$$f: [-1, 3] \rightarrow [-8, 72]$$

$$f(x) = 4x^3 - 12x$$

$$f(-1) = 8 \quad f(3) = 72$$

$$f(x) = 12x^2 - 12$$



$$f(1) = -8$$

19. Ans (C)

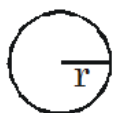
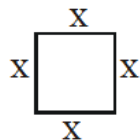
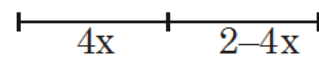
$$(x-3)^2 + (y-4)^2 = 1$$

$$(x-4)^2 + (y-3)^2 = 1$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$\Rightarrow$ circles are ortho-gonal

20. Ans (A)



$$2\pi r = 2 - 4x$$

$$A = \pi r^2 + x^2$$

$$A = \pi \left(\frac{2-4x}{2\pi} \right)^2 + x^2$$

$$\frac{dA}{dx} = -\frac{1}{\pi} 4(1-2x) + 2x$$

$$\frac{dA}{dx} = 0 \Rightarrow x = \frac{2}{\pi+4}$$

$$\frac{d^2A}{dx^2} > 0$$

MATHEMATICS

SEC-2

1. Ans (1.50)

$$(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} = \frac{1}{2} \vec{b} + \frac{1}{2} \vec{c}$$

$\vec{b}$ and $\vec{c}$ are non collinear so

$$\vec{a} \cdot \vec{c} = \frac{1}{2} \Rightarrow \cos \beta = \frac{1}{2} \Rightarrow \beta = 60^\circ$$

$$\vec{a} \cdot \vec{b} = -\frac{1}{2} \Rightarrow \cos \beta = -\frac{1}{2} \Rightarrow \alpha = 120^\circ$$

$$\cos(\alpha - \beta) - \cos(\alpha + \beta) = 2 \sin \alpha \sin \beta = \frac{3}{2}$$

2. Ans (0.33)

$$\text{Let } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$$

$$\frac{2b^2}{a} = 4 \Rightarrow b^2 = 2a$$

$$\text{Also } a - ae = \frac{3}{2} \quad b^2 = a^2(1 - e^2)$$

$$a(1 - e) = \frac{3}{2} \quad 2 = a(1 - e^2)$$

$$\Rightarrow \frac{1 - e^2}{1 - e} = \frac{4}{3} \Rightarrow e = \frac{1}{3}$$

3. Ans (2.00)

$$\sqrt{x} + \sqrt{y} = 3$$

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} y' = 0$$

$$y'_{(4,1)} = -\frac{1}{2}$$

$$\text{length of subtangent} = \left| \frac{y}{y'} \right| = 2$$

4. Ans (2.00)

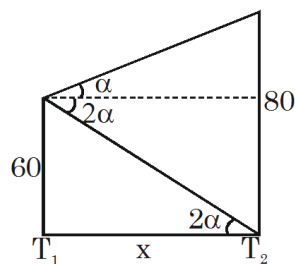
$$f(x) = x + \frac{1}{x^2}$$

$$f'(x) = 1 - \frac{2}{x^3} = \frac{x^3 - 2}{x^3}$$

$$f(x) \text{ is minimum at } x = 2^{1/3}$$

$$\alpha = 2^{1/3} \Rightarrow \alpha^3 = 2$$

5. Ans (34.64)



$$\tan 2\alpha = \frac{60}{x}$$

$$\tan \alpha = \frac{20}{x}$$

$$x = 20\sqrt{3}$$

6. Ans (2.50)

$$\lim_{x \rightarrow \sqrt{a}} \frac{b - \cos(x^2 - a)}{(x^2 - a) \sin(cx^2 - a)}; \left(\frac{0}{0} \right) \text{ if } b = 1$$

$$L = \lim_{x \rightarrow \sqrt{a}} \frac{1 - \cos(x^2 - a)}{(x^2 - a)^2} \cdot \frac{(x^2 - a)}{\sin(cx^2 - a)}, \left(\frac{0}{0} \right)$$

$$\text{if } c = 1$$

$$L = \frac{1}{2}$$

7. Ans (2.00)

$$\alpha + \beta + \gamma = 2$$

$$\Sigma \alpha \beta = 0$$

$$\alpha \beta \gamma = 1$$

$$\text{Also } \alpha^3 - 2\alpha^2 - 1 = 0 \Rightarrow \alpha^3 - 1 = 2\alpha^2$$

$$\frac{a_{11} - a_8}{a_{10}} = \frac{\Sigma \alpha^8 (\alpha^3 - 1)}{\Sigma \alpha^{10}} = \frac{2 \Sigma \alpha^{10}}{\Sigma \alpha^{10}} = 2$$

8. Ans (0.66 or 0.67)

$$f(x) = \frac{\int_0^{x^2} \sin \sqrt{x} dx}{x^3} \left(\frac{0}{0} \right)$$

$$f(x) = \frac{\sin x \cdot 2x}{3x^2} = \frac{2}{3}$$

9. Ans (8.66 or 8.67)

$$\sum_{i=1}^9 x_i = 80 \quad \sum_{i=1}^9 x_i^2 = 800$$

wrong observation = 3

correct observation = 4

$$\sum_{i=1}^9 (x_i)_{\text{new}} = 81, \bar{x}_{\text{new}} = 9, \sum_{i=1}^9 (x_i^2)_{\text{new}} = 807$$

corrected variance =

$$\frac{\sum (x_i)_{\text{new}}^2}{9} - (\bar{x}_{\text{new}})^2 = \frac{78}{9}$$

10. Ans (50.00)

$$T_{r+1} = {}^{60}C_r 3^{\frac{60-r}{2}} 5^{\frac{r}{3}}$$

(for rational terms) r is multiple of 6

$$r = 0, 6, 12, \dots, 60$$

Number of rational terms = 11